

```
In [1]: import pandas as pd

chicks = pd.read_csv("chickwts.csv")

chicks.groupby("feed")["weight"].mean().so
```

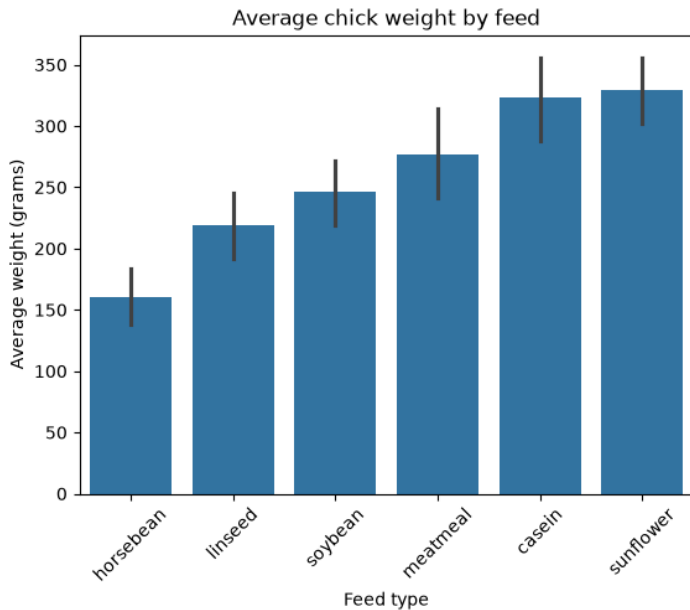
```
Out[1]: feed
horsebean    160.200000
linseed      218.750000
soybean      246.428571
meatmeal     276.909091
casein       323.583333
sunflower    328.916667
Name: weight, dtype: float64
```

```
In [2]: import seaborn as sns
import matplotlib.pyplot as plt

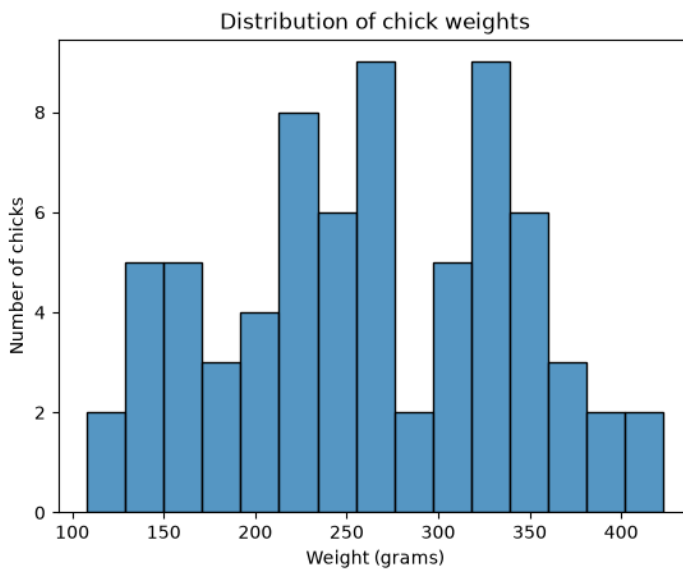
sns.barplot(
    data=chicks,
    x="feed",
    y="weight",
    order=chicks.groupby("feed")["weight"]
)

plt.title("Average chick weight by feed")
plt.xlabel("Feed type")
plt.ylabel("Average weight (grams)")
plt.xticks(rotation=45)

plt.show()
```



```
In [3]: sns.histplot(data=chicks, x="weight", bins  
plt.title("Distribution of chick weights")  
plt.xlabel("Weight (grams)")  
plt.ylabel("Number of chicks")  
plt.show())
```



```
In [4]: museum = pd.read_csv("museum_visitors.csv")  
        museum.head()
```

```
Out[4]: <style scoped> .dataframe tbody tr th:only-of-type {
vertical-align: middle; } .dataframe tbody tr th {
vertical-align: top; } .dataframe thead th { text-align:
right; } </style> <thead> <th></th> <th>Date</th>
<th>Avila Adobe</th> <th>Firehouse Museum</th>
<th>Chinese American Museum</th> <th>America
Tropical Interpretive Center</th> </thead> <tbody>
<th>0</th> <th>1</th> <th>2</th> <th>3</th>
<th>4</th> </tbody> <tbody></tbody>
```

2014-01-01	24778	4486	1581	6602
------------	-------	------	------	------

2014-02-01	18976	4172	1785	5029
------------	-------	------	------	------

2014-03-01	25231	7082	3229	8129
------------	-------	------	------	------

2014-04-01	26989	6756	2129	2824
------------	-------	------	------	------

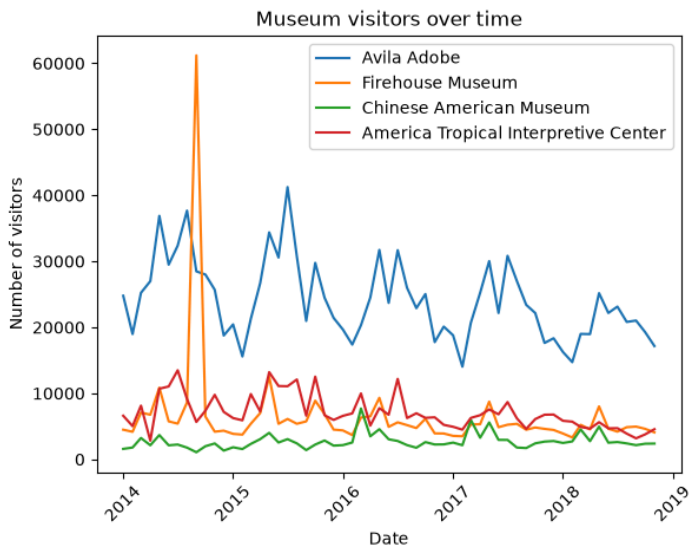
2014-05-01	36883	10858	3676	10694
------------	-------	-------	------	-------

```
In [5]: museum["Date"] = pd.to_datetime(museum["Da

sns.lineplot(data=museum, x="Date", y="Avi
sns.lineplot(data=museum, x="Date", y="Fir
sns.lineplot(data=museum, x="Date", y="Chi
sns.lineplot(data=museum, x="Date", y="Ame
            label="America Tropical Inter

plt.title("Museum visitors over time")
plt.xlabel("Date")
plt.ylabel("Number of visitors")
plt.legend()
plt.xticks(rotation=45)

plt.show()
```



<p>Avila adobe museum has a sitable visitors generally has the higher visitors over time .except the perioed 2014and 2015 the firehouse museum has the highest visitors.</p>

```
In [6]: sales = pd.read_csv("sales_clean.csv")
        sales.head()
```

```
Out[6]: <style scoped> .dataframe tbody tr th:only-of-type {
vertical-align: middle; } .dataframe tbody tr th {
vertical-align: top; } .dataframe thead th { text-align:
right; } </style> <thead> <th></th> <th>order_id</th>
<th>date</th> <th>product</th> <th>price</th>
<th>qty</th> <th>zip</th> </thead> <tbody>
<th>0</th> <th>1</th> <th>2</th> <th>3</th>
<th>4</th> </tbody> <tbody></tbody>
```

1254	2026-04-06	mug	11.36	4	60614
------	------------	-----	-------	---	-------

1057	2026-05-30	charger	14.79	3	30303
------	------------	---------	-------	---	-------

1150	2026-05-06	mug	5.50	1	10001
------	------------	-----	------	---	-------

1066	2026-05-30	notebook	37.53	1	98101
------	------------	----------	-------	---	-------

1114	2026-04-06	webcam	45.59	4	90405
------	------------	--------	-------	---	-------

```
In [7]: sales.columns
```

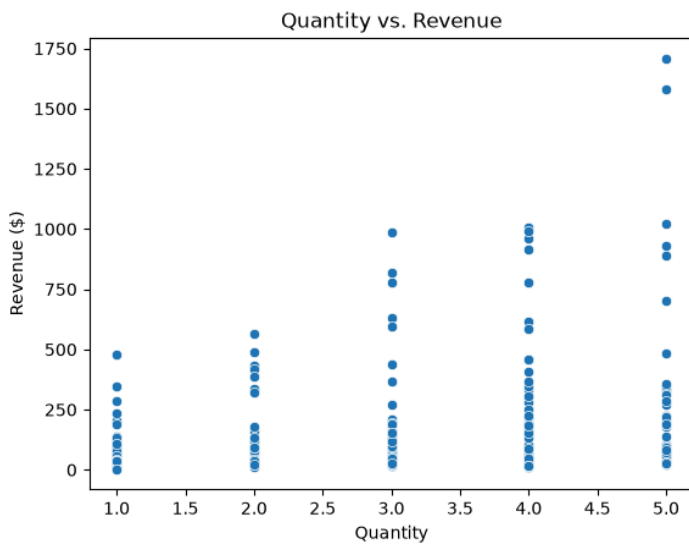
```
Out[7]: Index(['order_id', 'date', 'product', 'pr
ice', 'qty', 'zip'], dtype='str')
```

```
In [8]: sales["revenue"] = sales["price"] * sales[
sales.head()]
```

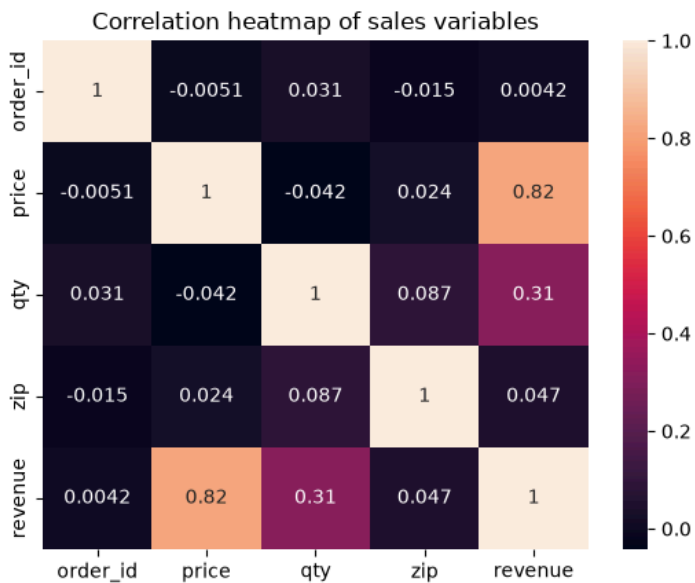
```
Out[8]: <style scoped> .dataframe tbody tr th:only-of-type {
vertical-align: middle; } .dataframe tbody tr th {
vertical-align: top; } .dataframe thead th { text-align:
right; } </style> <thead> <th></th> <th>order_id</th>
<th>date</th> <th>product</th> <th>price</th>
<th>qty</th> <th>zip</th> <th>revenue</th>
</thead> <tbody> <th>0</th> <th>1</th> <th>2</th>
<th>3</th> <th>4</th> </tbody> <tbody></tbody>
```

1254	2026-04-06	mug	11.36	4	60614	45.44
1057	2026-05-30	charger	14.79	3	30303	44.37
1150	2026-05-06	mug	5.50	1	10001	5.50
1066	2026-05-30	notebook	37.53	1	98101	37.53
1114	2026-04-06	webcam	45.59	4	90405	182.36

```
In [9]: sns.scatterplot(data=sales, x="qty", y="re
plt.title("Quantity vs. Revenue")
plt.xlabel("Quantity")
plt.ylabel("Revenue ($)")
plt.show()
```



```
In [10]: sns.heatmap(sales.corr(numeric_only=True),  
plt.title("Correlation heatmap of sales va  
plt.show()
```



<p>the null hypothesis roughly all the chicken between 250 to 260 gram.</p>

```
In [11]: from scipy.stats import ttest_ind

casein = chicks[chicks["feed"] == "casein"]
horsebean = chicks[chicks["feed"] == "horsebean"]

ttest_ind(casein, horsebean)
```

```
Out[11]: TtestResult(statistic=np.float64(7.019741040397541), pvalue=np.float64(8.254541016953185e-07), df=np.float64(20.0))
```

<p>The conclusion Casein-fed chicks were on average 163.38 grams heavier than horsebean-fed chicks ($p < 0.001$), so we reject the null hypothesis. However, this result shouldn't be accurate because it is short list of data</p>

Honest-chart audit I used clear labels for the dates and museum names so it is easy to see which line belongs to each museum. I also labeled the y-axis as "Number of visitors" so the measurement is clear.

In []: